

# No causality gap for atomic capacity on a locally finite infinite tree

Research note developed with Jan Mikulík

10 October 2026

## Abstract

For the transcript laws of all finite-horizon deterministic policies on a locally finite controlled tree, define  $a_K$  to be the smallest mass carried by the  $K$  most likely transcripts, uniformly over policies and horizons. We prove that  $q = \lim_K a_K$  is exactly the maximum total atomic mass of an exact causal representation. The maximum is attained and equals the corresponding maximum without causality constraints. Thus the unavoidable nonatomic mass is exactly  $1 - q$ . The proof uses finite causal greedy extraction, a quantitative concentration estimate, and a diagonal limit with an explicitly realized residual flow. A further compactness argument shows that bounded finite-horizon optimal entropies give one optimal countable seed for the entire infinite tree, with no extra entropy cost.

**Status and scope.** This is a mathematical proof conditional on the finite causal extraction and minimax theorem of the finite-Shannon-theorem project. It is not a Lean certificate, and publication priority is not claimed. Infinite families of ordinary couplings and their uniform concentration condition are already treated by Li [2]. The assertion examined here is attainment of the *same atomic capacity by compatible strategies*, including  $q < 1$ . This does not determine the finite Shannon entropy gap  $G_*$ .

## 1 Objects and statement

Let  $T$  be a rooted controlled history tree. At every finite history  $h$  there are finitely many available actions, each with a finite nonempty output alphabet and a prescribed probability kernel. We assume the alphabets are finite at every node, not uniformly bounded. Terminal nodes can be padded by a forced action and output. Zero-probability outputs may be deleted; deterministic singleton alphabets are permitted.

Write  $p(h)$  for the product of output probabilities along  $h$ . For every available action  $a$ ,

$$p(\emptyset) = 1, \quad \sum_y p(hay) = p(h).$$

At each finite depth there are finitely many histories. A deterministic response strategy selects one output at every history-action pair. The space  $\mathcal{S}$  of all such strategies is a countable product of finite discrete spaces, hence compact and metrizable. For a history  $h$ , let  $E_h \subseteq \mathcal{S}$  be the set of strategies agreeing with its output choices. This is a clopen cylinder, and

$$E_h = \bigsqcup_y E_{hay} \quad \text{for every action } a.$$

A Borel probability measure  $\mu$  on  $\mathcal{S}$  is an exact causal representation if  $\mu(E_h) = p(h)$  for every finite history  $h$ . This is equivalent to exactness for every deterministic adaptive policy at every finite horizon, and therefore for the corresponding infinite path laws.

Let  $P_{n,\pi}$  be the finite transcript law at horizon  $n$  under policy  $\pi$ . Put

$$A_K(P) = \max_{D: |D| \leq K} P(D), \quad (1)$$

$$a_K(T) = \inf_{n,\pi} A_K(P_{n,\pi}), \quad q(T) = \lim_{K \rightarrow \infty} a_K(T). \quad (2)$$

The limit exists by monotonicity and belongs to  $[0, 1]$ . For a Borel probability on a compact metrizable space define its total atomic mass by

$$\text{At}(\mu) = \sum_{s: \mu(\{s\}) > 0} \mu(\{s\}).$$

There are at most countably many terms.

Ordinary couplings below are Borel probability laws on the countable product of the transcript spaces indexed by all pairs  $(n, \pi)$ , with the specified coordinate marginals. They need not be compatible with a single strategy or even consistent between horizons.

**Theorem 1** (Atomic capacity). *For every such tree,*

$$\boxed{\max_{\mu \text{ exact causal}} \text{At}(\mu) = \max_{\Gamma \text{ ordinary}} \text{At}(\Gamma) = q(T).}$$

*Both maxima are attained. In particular, an exact countably supported causal seed for the entire infinite tree exists if and only if  $q(T) = 1$ . If  $q(T) < 1$ , every exact causal representation has nonatomic mass at least  $1 - q(T)$ , and a representation attaining this minimum exists.*

## 2 The finite concentration estimate

We use the following part of the finite theorem [1]. After a finite number of causal greedy extractions, the residual cylinder masses form a nonnegative controlled flow. If its root mass is  $R$ , its next extractable strategy weight  $w$  equals the smallest, over policies, of the largest residual transcript mass. Extraction terminates on each finite tree and gives an exact finite seed.

**Lemma 2.** *Suppose every policy on a finite controlled tree has  $K$  transcripts carrying total mass at least  $a$ . The first  $M$  greedy atoms, padded with zeros if extraction has ended, have total mass at least*

$$a[1 - (1 - 1/K)^M].$$

*Consequently the same is true of the  $M$  largest seed atoms.*

*Proof.* Let  $R_m$  be the residual root mass after  $m$  steps, with  $R_0 = 1$ . The minimax identity supplies a policy all of whose residual transcript masses are at most the next weight  $w_{m+1}$ . On this policy select at most  $K$  original transcripts of mass at least  $a$ . The residual mass outside this set is at most  $1 - a$ , so

$$R_m - (1 - a) \leq K w_{m+1}.$$

Since  $R_{m+1} = R_m - w_{m+1}$ ,

$$R_{m+1} - (1 - a) \leq (1 - 1/K)(R_m - (1 - a)).$$

Iteration gives  $R_M \leq 1 - a + a(1 - 1/K)^M$ . The same bound holds after termination. Subtract from one. For  $K = 1$  and  $M \geq 1$  the displayed formula has its usual value  $a$ .  $\square$

### 3 Proof of atomic capacity

*Proof of Theorem 1. Upper bound, even without causality.* Take any  $m$  atoms of an ordinary coupling, of total mass  $b$ . Their images in any one transcript coordinate lie in at most  $m$  transcripts, so  $b \leq A_m(P_{n,\pi})$  for every  $(n, \pi)$ . Hence  $b \leq a_m \leq q$ . Exhausting all its atoms gives  $\text{At}(\Gamma) \leq q$ . The same argument applies directly to a causal representation by projecting each atom through a policy.

*Limits of weights and strategies.* For each horizon  $n$ , take a finite greedy causal seed, order its weights as  $w_1^{(n)} \geq w_2^{(n)} \geq \dots \geq 0$ , and pad by zeros. Extend each finite strategy arbitrarily to an element  $s_i^{(n)}$  of  $\mathcal{S}$ . By compactness and diagonal selection there is one subsequence  $n_j \rightarrow \infty$  such that, for every  $i$ ,

$$w_i^{(n_j)} \rightarrow w_i, \quad s_i^{(n_j)} \rightarrow s_i.$$

Set  $\nu = \sum_i w_i \delta_{s_i}$  and  $A = \sum_i w_i \leq 1$ ; repeated limit strategies are simply merged as measures. For fixed  $K, M$ , Lemma 2, with  $a = a_K(T)$ , gives

$$\sum_{i=1}^M w_i \geq a_K(T) [1 - (1 - 1/K)^M].$$

First let  $M \rightarrow \infty$  and then  $K \rightarrow \infty$ . It follows that  $A \geq q$ .

*No invalid passage of infinitely many atoms through a limit.* For fixed  $h$  and  $M$ , convergence of strategies makes membership in the clopen set  $E_h$  eventually constant for each of the first  $M$  indices. For large  $j$ , the finite seed is exact at  $h$ . Thus

$$\sum_{i=1}^M w_i \mathbf{1}_{E_h}(s_i) = \lim_j \sum_{i=1}^M w_i^{(n_j)} \mathbf{1}_{E_h}(s_i^{(n_j)}) \leq p(h).$$

Monotone convergence in  $M$  proves  $\nu(E_h) \leq p(h)$ . Define

$$r(h) = p(h) - \nu(E_h) \geq 0.$$

Since the  $E_{hay}$  partition  $E_h$  for each action,  $\sum_y r(hay) = r(h)$  and  $r(\emptyset) = 1 - A$ .

*Realization of the missing mass.* If  $A < 1$ , assign, independently at every history–action pair, an output with probabilities  $r(hay)/r(h)$  whenever  $r(h) > 0$ , and any distribution when  $r(h) = 0$ . The countable product law  $\lambda$  of these response choices satisfies

$$\lambda(E_h) = r(h)/(1 - A).$$

Indeed this follows by induction along  $h$  from the displayed ratios; after a zero residual prefix both sides vanish. Hence  $\mu = \nu + (1 - A)\lambda$  is exact. If  $A = 1$ , simply set  $\mu = \nu$ . As  $\nu$  is atomic of mass  $A$ ,  $\text{At}(\mu) \geq A \geq q$ . The upper bound already proved gives  $\text{At}(\mu) = A = q$ . In particular, the residual law in this construction has no atoms when  $q < 1$ .

Finally map  $s \in \mathcal{S}$  to its full tuple of finite-policy transcripts. This is a measurable map into the countable product of finite spaces. The image of  $\mu$  is an ordinary coupling and has atomic mass at least  $q$ , since images of atoms remain atoms. The upper bound makes its atomic mass exactly  $q$ . This proves attainment on both sides.  $\square$

### 4 Information spectrum and entropy

All logarithms in this section have base two. Define

$$F_n(t) = \min_{\pi} P_{n,\pi} \{-\log P_{n,\pi}(X) \leq t\}, \quad F_{\infty}(t) = \inf_n F_n(t), \quad t \geq 0.$$

Refining a transcript cannot reduce its information, and policies extend to longer horizons. Therefore  $F_n(t)$  decreases with  $n$ . For an individual finite law  $P$ , writing its information CDF as  $F_P$ ,

$$F_P(t) \leq A_{\lfloor 2^t \rfloor}(P), \quad A_K(P) \leq F_P(t) + K2^{-t}.$$

Taking infima over all finite-policy laws proves

$$\boxed{q(T) = \lim_{t \rightarrow \infty} F_\infty(t).}$$

Thus atomic capacity is precisely the mass that does not escape to infinite information in the uniform policy envelope. This statement does not require treating a possibly defective  $F_\infty$  as a probability CDF.

Let  $C_n$  be the minimum entropy of an exact seed for the depth- $n$  tree. Let  $C_\infty$  be the infimum entropy among countably supported exact seeds for the full tree, with value  $+\infty$  if there is no such seed. Set

$$L_n = \int_0^\infty (1 - F_n(t)) dt, \quad L_\infty = \sup_n L_n, \quad c_* = (1 + \log e)/2.$$

**Theorem 3** (No extra entropy at the projective limit). *In the extended real numbers,*

$$\boxed{C_\infty = \sup_n C_n.}$$

*If the right-hand side is finite, the infimum on the left is attained. Consequently the finite Shannon theorem implies*

$$L_\infty \leq C_\infty \leq L_\infty + c_*.$$

*In particular,  $L_\infty < \infty$  is equivalent to the existence of one finite-entropy exact causal seed for the entire infinite tree.*

*Proof.* Restriction of a seed proves  $C_n \leq C_{n+1} \leq C_\infty$ . If  $B = \sup_n C_n = +\infty$ , this already proves the equality. Assume  $B < \infty$  and choose a minimizing finite seed at each horizon. Such minimizers exist: one can minimize entropy over the compact finite-dimensional polytope of laws on finite strategies, merging identical strategies if necessary. Order and pad weights as above. For  $i > K$ ,  $w_i^{(n)} \leq 1/(K+1)$ , whence

$$\sum_{i>K} w_i^{(n)} \leq \frac{B}{\log(K+1)}. \quad (*)$$

Diagonal compactness gives limiting weights and strategies. The uniform tail bound (\*) forces  $\sum_i w_i = 1$ . The cylinder argument in the previous proof gives  $\nu(E_h) \leq p(h)$ . Since both are probability flows, equality follows at each depth by summing over the leaves of any policy following  $h$ , or by passing the uniformly small tails directly through the limit.

The countable label seed  $i$  with masses  $w_i$  realizes these strategies. With  $0 \log 0 = 0$ , continuity of each summand and Fatou's lemma give

$$H(w) = \sum_i -w_i \log w_i \leq \liminf_j H(w^{(n_j)}) = B.$$

Together with restriction this yields  $C_\infty = B$ , with attainment. Finally the finite theorem gives  $L_n \leq C_n \leq L_n + c_*$ ; take suprema. No continuity of Shannon entropy on an unrestricted countable simplex is being assumed.  $\square$

**Three regimes.** If  $q < 1$ , every exact representation has an unavoidable nonatomic component. If  $q = 1$  but  $L_\infty = \infty$ , a countable exact seed exists but all such seeds have infinite entropy. If  $L_\infty < \infty$ , an optimal finite-entropy countable seed exists. Countable support is not the same as finite support.

## 5 A strict quantifier separation

Consider a deterministic waiting chain. At waiting stage  $m \geq 1$  the controller can either wait for the next stage or trigger an experiment. Triggering at stage  $m$  produces  $m$  independent fair bits and then stops; there are no choices during the experiment. A policy that never triggers has a deterministic infinite transcript. The tree is locally finite, with at most two actions and at most two outputs at every node.

Every individual deterministic policy therefore has a finite-support full-path law: either one path, or  $2^m$  paths for some finite  $m$ . Nevertheless, for each fixed  $K$ , choosing arbitrarily large  $m$  yields

$$a_K \leq K2^{-m}, \quad a_K = 0, \quad q = 0.$$

Hence *every exact common representation is atomless*, even without a causality requirement. An elementary direct proof is that an atom of mass  $w > 0$  would have to map into a transcript of mass  $2^{-m} < w$  for a sufficiently large triggered experiment. This example illustrates why separate finite-support simulations do not imply one discrete seed. It is not claimed as a new counterexample in the general theory of ordinary coupling.

For any prescribed  $q_0 \in (0, 1)$ , first choose a deterministic absorbing branch with probability  $q_0$  and the waiting construction with probability  $1 - q_0$ . Then  $a_K = q_0$  for every  $K \geq 1$ : the absorbing transcript gives the lower bound, and triggering an arbitrarily long experiment gives the upper bound. Thus the full interval of atomic capacities is realized even with binary branching.

**What the result does not say.** The diagonal argument is an existence proof, not a uniform algorithm for an arbitrary computable infinite tree. It does not bound the time needed to sample the seed, determine  $G_*$ , prove a new asymptotic coefficient, or assert that this infinite extension has been formally verified. Its structural conclusion is the exact equality of ordinary and causal atomic capacities, including attainment and partial atomic mass, and the absence of a further entropy charge when finite-horizon optima stay bounded.

## References

- [1] J. Mikulík, *An exact causal seed on a finite controlled tree*, finite-Shannon-theorem repository. The finite minimax/extraction and Shannon statements are the inputs used here. <https://github.com/jendamikulik/finite-Shannon-theorem>.
- [2] C. T. Li, *Efficient Approximate Minimum Entropy Coupling of Multiple Probability Distributions*, IEEE Transactions on Information Theory 67(8), 5259–5268, 2021. <https://arxiv.org/abs/2006.07955>. Definition 6 and Corollary 9 treat uniform concentration and ordinary discrete couplings for infinite families; this prior work must be distinguished from causal compatibility on a history tree.